

**DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY/
MANAGEMENT/ COMMERCIAL PRACTICE.
CLIMATE RESPONSIVE ARCHITECTURE**

Time: 3 hours

Maximum Marks: 75

PART A**I. Answer all the following questions in one word or sentence.****(9 x 1 = 9 Marks)**

Module Outcome Cognitive level

1	What is called Relative Humidity?	M 1.01	R
2	Define Site Climate	M 1.03	R
3	Define thermal balance equation.	M2.03	R
4	Differentiate between horizontal and vertical shading devices	M2.02	U
5	Define physiological cooling.	M3.01	R
6	Name any 2 thermal insulation methods.	M3.01	R
7	Define Roof ponding.	M3.01	R
8	Name the type of planning followed in hot-dry climates.	M4.01	R
9	Double roof construction is preferred in _____ climate.	M4.01	R

PART B**II. Answer any eight questions from the following****(8 x 3 = 24 Marks)**

Module Outcome Cognitive level

1	Differentiate between Climate and Weather	M 1.01	U
2	Differentiate between Macro and Micro climate	M 1.01	U

3	Explain the factors affecting site climate.	M 1.01	U
4	Classify Tropical climates and its sub groups.	M1.01	R
5	Explain human body's heat regulatory mechanism.	M2.03	U
6	What are the factors affecting air flow around the building?	M3.01	R
7	What is called Stack effect?	M3.01	R
8	What is called Passive Cooling. Give any 3 examples.	M3.01	R
9	Illustrate and explain the working of Solar Chimney.	M3.01	A
10	Explain the form and planning of buildings in warm humid climate.	M4.01	U

PART C

III. Answer all questions from the following

(6 x 7 = 42 Marks)

Module Outcome Cognitive level

1	Explain Global wind pattern	M1.01	U
	OR		
2	Explain Thermal balance of earth.	M1.01	U
3	Explain the heat exchange process in buildings.	M2.01	U
	OR		
4	Explain Greenhouse effect and different methods to reduce solar heat gain through windows.	M2.03	U
5	What is meant by Structural controls? What is the importance of Structural control in building design?	M2.02	R
	OR		
6	What are called thermal comfort factors? What are the subjective variables affecting human comfort?	M2.03	R
7	<i>Wind scoops are a good choice to be part of an HVAC system as a source of cooling and ventilation for a building. Substantiate this statement with sketches.</i>	M3.02	A
	OR		
8	Illustrate and explain the differences between Earth coupling and Earth air tunnel	M3.02	A

9	Explain the planning features of traditional shelters in Composite climates.	M4.01	U
	OR		
10	Explain the planning features of traditional shelters in Warm humid climate.	M4.01	U
11	Explain the features of external spaces, roof and walls in Hot dry climate.	M4.01	U
	OR		
12	Explain the features of external spaces, roof and walls in Tropical upland climate.	M4.02	U

